

Homework 4

Name: Solution key

Problem 1 (15 pts)

We solve this problem by appealing to the subspace embedding theorem proven in class. All vectors with a particular sparsity pattern lie in a k dimensional linear subspace – they can be written as the linear combination of k particular standard basis vectors. Accordingly, to preserve the norm of all vectors with a particular sparsity pattern, we would need a JL embedding with:

$$O\left(\frac{k + \log(1/\delta)}{\epsilon^2}\right) \text{ rows.}$$

There are $\binom{d}{k} = O(d^k)$ possible sparsity patterns for a d dimensional vector. So, to preserve the norm of all sparse vectors (i.e. in any of these $O(d^k)$ possible subspaces) with probability say $1/d$ (high probability), we can set $\delta = d/d^k$. Accordingly, we need to set:

$$O\left(\frac{k + \log(d^k/d)}{\epsilon^2}\right) = O\left(\frac{k \log d}{\epsilon^2}\right).$$

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Problem 2 (15 pts)

The proof is by contradiction.

Suppose there are two vectors $x \neq x'$ that are both k -sparse and have $Ax = Ax'$. Then $z = x - x'$ is a $2k$ -sparse vector which satisfies $Az = \vec{0}$. If $x \neq x'$ then $z \neq 0$.

Let $\mathcal{I} \subset \{1, \dots, n\}$ be some set of $2k$ indices which contains all non-zero indices in z (of which there are at most $2k$). Let $A_{\mathcal{I}} \in \mathbb{R}^{2k \times 2k}$ be a matrix containing the columns of A corresponding to the indices in $\mathcal{I}$. Similarly, let $z_{\mathcal{I}} \in \mathbb{R}^{2k}$ contain all non-zero entries in z .

$A_{\mathcal{I}} z_{\mathcal{I}} = \vec{0}$ and since $z_{\mathcal{I}}$ is non-zero, it follows that there must be some non-zero, length $2k$ vector w such that $w A_{\mathcal{I}} = 0$. This can be proven in a variety of ways. In particular, since $A_{\mathcal{I}}$ has $2k$ columns, $A_{\mathcal{I}} z_{\mathcal{I}} = \vec{0}$ implies that $A_{\mathcal{I}}$ has column rank $\leq 2k - 1$. So it has row rank $\leq 2k - 1$. But since it has $2k$ rows, these rows cannot be linearly independent, which implies the existence of w .

Let $p_w(x)$ be the polynomial $p_w(x) = w_1 + w_2 x + w_3 x^2 + \dots + w_{2k} x^{2k-1}$. Since w is non-zero, p_w is not the zero polynomial. But, if $w A_{\mathcal{I}} = 0$, then $p_w(a_i) = 0$ for all $i \in \mathcal{I}$. This would lead to a contradiction, because it would imply that $p_w(x)$ has $2k$ roots, even though it only has degree $2k$.

We conclude that there cannot be distinct vectors x, x' with $Ax = Ax'$. And thus, at least information theoretically, it is possible to recover x from Ax .

Problem 3 (10 pts)

The idea is to let B equal:

$$B = \begin{bmatrix} AD_0 \\ AD_1 \\ AD_2 \\ \dots \\ AD_s \end{bmatrix}$$

where $s = \lfloor \log n + 1 \rfloor$. The parameters (i.e. number of rows in A) will be chosen later. For now, we just specify that D_0 is the identity matrix and each D_j for $j > 0$ is a randomly chosen diagonal 0,1 matrix with $n/2^j$ entries equal to 1. In particular, choose D_1 to have a 1 at $n/2$ locations chosen uniformly at random. Let D_2 have a 1 at $n/4$ locations chosen uniformly at random from those in D_1 , so on and so forth. So $AD_s = 0$.

Our output Bx contains $AD_0x, AD_1x, \dots, AD_sx$.

If x is 1-sparse, then (i, x_i) can be recovered from Ax . If it is not sparse, then we will show that we can recover (i, x_i) from a lower level AD_jx .

Specifically, to recover an entry of x , run our recovery algorithm on $y_0 = AD_0x, y_1 = AD_1x, \dots, y_s = AD_sx$. Look at the lowest level (i.e. largest j) such that $\text{decode}(y_j)$ returns a non-zero vector $\tilde{x}_j$. Return any index/value pair in $\tilde{x}_j$.

It's not hard to see that, if we choose A to have k rows, this recovery strategy will succeed in returning a correct answer as long as there exists some j for which $1 \leq \|D_jx\|_0 \leq k$.

We want to show that this happens with good probability for $k = O(1)$. In particular, consider the $w \in 0, 1, \dots$ such that $2^w \leq \|x\|_0 \leq 2^{w+1}$. We only consider level D_{w-c} for some constant c .

We first show that $\Pr[\|D_{w-c}x\|_0 = 0] \leq 1/e^c$. Specifically, the probability that any particular non-zero entry in x is non-zero in $D_{w-c}x$ is $1/2^{w-c}$. The probability that at least one entry is non-zero is:

$$\Pr[\|D_{w-c}x\|_0 = 0] \leq (1 - 1/2^{w-c})^{\|x\|_0} \leq (1 - 1/2^{w-c})^{2^w} \leq (1/e)^c.$$

At the same time, $\Pr[\|D_{w-c}x\|_0 \geq 2^{2^{c-1}}] \leq 1/2^c$. This follows from Markov's inequality because $\mathbb{E}[\|D_{w-c}x\|_0] = \frac{\|x\|_0}{2^{w-c}} \leq \frac{2^{w+1}}{2^{w-c}} = 2^{c+1}$.

So by a union bound, with probability $1 - 1/e^c - 1/2^c$, we have $1 \leq \|D_{w-c}x\|_0 \leq 2^{c-1}$. Setting $c = O(1)$, with any constant success probability, we have $1 \leq \|D_{w-c}x\|_0 \leq O(1)$.

Problem 4 (10 pts)

Jesse and Leslie construct a sensing matrix A as in Problem 1 with $q_1, \dots, q_n = 1, \dots, n$. They then construct a random B as in Problem 2, using the same exact random bits. So they both hold the same sensing matrix B .

Then Jesse constructs an n length binary vector v which has a 1 at all indices $i \in S_B$. Similarly, Leslie constructs an n length binary vector v' which has a 1 at all indices $i \in S_A$.

Jesse computes Bv . This is an $O(\log n)$ length vector. The magnitude of each entry is bounded by $n \cdot n^{O(1)}$, so can be represented with $O(\log n)$ bits. So she can send Bv to Leslie while using only $O(\log^2 n)$ bits.

Upon receiving Bv , Leslie computes $Bv - Bv' = B(v - v')$. She runs the recovery algorithm for Problem 2 to extract any non-zero index in $(v - v')$, which must correspond to some entry Leslie and Jessie do not have in common.